Saffron Aqueous Extract Prevents Metabolic Syndrome in Patients with Schizophrenia on Olanzapine Treatment: A Randomized Triple Blind Placebo Controlled Study

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Key words
- saffron
- olanzapine
- FBS
- metabolic syndrome
- insulin resistance

Abstract

Objective: The aim of this study was to assess whether saffron aqueous extract (*SAE*) or its active constituent, crocin, prevents olanzapine-induced metabolic syndrome (MetS) and insulin resistance in patients with schizophrenia.

Methods: 66 patients diagnosed with schizophrenia who were on olanzapine treatment (5–20 mg daily) were randomly allocated to receive a capsule of SAE (n = 22; 30 mg daily), crocin (n = 22; 30 mg daily) or placebo (n = 22) in a 12-week triple-blind trial. Patients were screened not to have MetS at baseline and further assessment was done at weeks 6 and 12. Measurement of fasting blood glucose (FBS) and serum lipids were repeated at weeks 2, 6 and 12. Fasting blood levels of insulin and HbA1c were also measured at baseline and week 12. HOMA-IR and HOMA-β were determined to evaluate insulin resistance.

Results: 61 patients completed the trial and no serious adverse effects were reported. Time-treatment interaction showed a significant difference in FBS in both SAE and crocin groups compared to placebo ($p = 0.004$). In addition, SAE could effectively prevent reaching the criteria of metabolic syndrome (0 patients) compared to crocin (9.1%) and placebo (27.3%) as early as week 6.

Conclusion: SAE could prevent metabolic syndrome compared to crocin and placebo. Furthermore, both SAE and crocin prevented increases in blood glucose during the study.

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Bibliography
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Introduction

Metabolic syndrome (MetS) is a serious problem and highly prevalent in patients with schizophrenia. It is interrelated to a constellation of several risk factors associated with hyperglycemia, triglyceridemia, decrease in HDL and cardio metabolic risk [1–3]. The most important factor for this excess risk is obesity accompanied with dyslipidemia, insulin resistance and hypertension [4].

Although a high prevalence of obesity and diabetes in patients with schizophrenia had been described long before the introduction of neuroleptic treatment, a contributing role for some second-generation antipsychotics has been described. There is emerging data that second-generation antipsychotics (such as olanzapine, clozapine, quetiapine and risperidone) [5] are associated with the development of metabolic syndrome because they have significant adverse effects on weight, lipids and glucose metabolism [6–8]. Glucose abnormalities have been shown to occur soon after initiation of treatment and are amendable after discontinuation of medication [6, 9–11]. Over recent years it has become apparent that olanzapine can have a negative impact on some of the modifiable risk factors of MetS [12, 13]. Melkersson et al. showed that olanzapine treatment was associated with weight gain and elevated levels of insulin and blood lipids as well as insulin resistance [14]. This effect could be prevented by a regular control of energy intake [15], but not all patients with schizophrenia are able to obey a structured life style. Despite the high burden of such metabolic changes, there is still no standard of care for preventing metabolic adverse effects of antipsychotic medications in patients with schizophrenia.

Herbals, which are traditionally used for the treatment of several medical conditions, are noticed for their possible use in preventing and treating metabolic syndrome [16]. This has led to a growing interest in using herbal products in psychiatric settings in recent years.

Saffron (*Crocus sativus L.*) and its active constitute crocin is one of the world's most researched spices, not only in vitro [18], but also in animal

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models [17, 19] and human disorders, so is introduced as an effective medicine [20, 21]. Apart from its value as a herbal remedy in traditional medicine and as a food additive [22, 23], recent studies point out its potential anti-diabetic effect and lipid profile enhancer in animal studies [17, 19, 23]. Our previous studies have shown hypoglycemic effects of both SAE and crocin, which seem to be a consequence of improved insulin secretion and reduced insulin resistance [17, 19]. Based on these results, in the present study, we investigated the effect of SAE and crocin to prevent olanzapine-induced metabolic syndrome and insulin resistance in patients with schizophrenia.

Patients and Methods

The present study was a 12-week randomized, triple-blind clinical controlled trial. The trial was conducted in the inpatient ward of Razi Psychiatric Hospital between March and September 2013. The proposal has been registered in 2 parts with the Iranian Registry of Clinical Trials (IRCT) (IRCT2013020312351N1 and IRCT2013081912351N2). The trial was performed in accordance with the Declaration of Helsinki and subsequent revisions and the procedure was approved by the ethical committee of University of Social Welfare and Rehabilitation Sciences, Tehran. In this study, written informed consent was obtained from all participants.

Participants

Male adult inpatients (18–65 years old) were included if they met the DSM-IV-TR criteria for schizophrenia [24] based on a structured clinical interview. Other inclusion criteria were: clinical stability without showing signs of the acute phase and negative history of previous olanzapine treatment.

Patients with acute general medical problems, coronary heart disease, metabolic syndrome, neurological disease, substance use problems (except for nicotine), treatment with anti-platelet medications or herbal medicines, intellectual disability and weight loss diets were excluded. Other exclusion criteria were: developing moderate to severe side effects, change in treatment regimen based on the attending psychiatrist's opinion and consent withdrawal. Those who left the study before completion were offered the standard care.

Preparation of saffron aqueous extract, crocin and placebo capsules

The saffron aqueous extract (SAE) and crocin were prepared using the method described in our previous reports [17, 19]. Identical capsules were filled with crocin or SAE. Each saffron capsule contained 15 mg dried SAE and each crocin capsule had 15 mg dried crocin. Placebo capsules were also filled with vehicle.

Measurements

Patients underwent a standard clinical assessment comprising a psychiatric evaluation using a structured diagnostic interview according to DSM-IV-TR [24], physical examination [including waist circumference and blood pressure measurement], a thorough medical history assessment, and screening lab tests for fasting blood sugar (FBS), lipid profile, fasting insulin and HbA1c. Then FBS and lipid profile were checked again in weeks 2, 6, and 12. Measurement of waist circumference and blood pressure was repeated at weeks 6 and 12. Fasting insulin and HbA1c were measured again in week 12. Insulin resistance and β-cell function assessment was measured by the HOMA model [25] using fasting values of both parameters: HOMA-IR = (glucose mg/dL * insulin (mU/L)/405) and HOMA-β = (360 * insulin mU/L/glucose mg/dL − 63) %,

The criteria for metabolic syndrome were adopted from the third adult treatment panel (ATP III) of the National Cholesterol Education Program (NCEP) with 3 or more of the following: (i) Abdominal obesity (waist circumference greater > 102 cm in men and > 88 cm in women), (ii) elevated triglycerides (≥ 150 mg/dL), (iii) decreased HDL cholesterol (< 40 mg/dL in men and < 50 mg/dL in women), (iv) hypertension (blood pressure ≥ 130/85 mm Hg) and (v) hyperglycemia (FBS ≥ 100 mg/dL) [26].

Study design

The antipsychotic regimen started with olanzapine 5 mg per day and could be increased up to 20 mg depending on the clinical response. After symptom resolution (4–6 weeks later, during the maintenance phase of the treatment), patients were randomized by computer generated codes. Allocation was concealed using sequentially numbered, opaque, sealed envelopes (SNOSE); the coding was only revealed after analysis of the results. Patients received either capsules of SAE (15 mg 2 times a day), crocin (15 mg 2 times a day) or placebo (similar capsules filled with vehicle, 2 times a day) as an adjunctive to olanzapine.

Side effects were systematically assessed and recorded throughout the study by a resident of psychiatry every other day during the first week and then on weeks 2, 6 and 12. This assessment started with an open ended question and then continued using a checklist including hypersensitivity, abnormal bleeding, GI disturbances and general satisfaction [27].

Statistical analysis

The sample size was estimated as 18 in each group according to Cohen Table, considering $a = 0.05$, $\beta = 0.2$, $S = 5$ and power = 80% [28]. Allowing for dropouts, 22 patients were recruited in each group.

A chi-square or one way ANOVA test evaluated differences in the baseline characteristics and the presence of metabolic syndrome between the 3 groups. A 2-way repeated measures analysis of variance (time-treatment interaction) with a 2-tailed post hoc LSD mean comparison test was also performed. The type of medication as a between-subjects factor (group) and time of measurements as the within-subjects factor (time) were considered in FBS and lipid profile, fasting insulin, HOMA models, and HbA1c parameter. Data are presented as mean (standard deviation) and differences were considered significant at $P < 0.05$.

Results

A total of 61 patients completed the study process. **Fig. 1** shows the process of the study. No serious adverse effects were seen.

Table 1 shows that there was no significant difference between patients randomly assigned to each group in terms of basic demographic data and dose of olanzapine and mean duration of morbidity. It should be mentioned that, in this study, level of education was calculated as the years spent in school or university.

Table 2, **3** describe components of metabolic syndrome and insulin resistance measured during the 12 week trial.

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As it can be seen in the tables, the repeated measurement analysis showed a significant difference between the 3 groups regarding levels of FBS (p = 0.004). Further post hoc LSD revealed that both SAE and crocin groups were different compared to the placebo group (p = 0.004), which means that SAE and crocin could prevent further increases in FBS. Furthermore, we included relevant baseline variables as covariate to this model. None of selected variables had a significant influence (p-values for age: 0.773, dose of olanzapine: 0.791, waist circumference: 0.604, HOMA-IR: 0.538, weight: 0.102, BMI: 0.594, HbA1c: 0.144) and the time-treatment interaction was still significant (p = 0.002), and both SAE and crocin groups were different compared to the placebo group.

SAE could also effectively prevent reaching the criteria of metabolic syndrome when compared to crocin and placebo; as 2 patients (9.1 %) from crocin group and 6 patients (27.3 %) from placebo group developed metabolic syndrome while none of the patients receiving SAE fulfilled the criteria for metabolic syndrome at week 12 (p = 0.022). This difference appeared as early as week 6 (p = 0.019). Results were still the same when considering age (p = 0.225) and dose of olanzapine (p = 0.665) as covariates and SAE could effectively prevent reaching the criteria of metabolic syndrome (p = 0.005).

Table 3 shows that the time-treatment interaction of components of insulin resistance was not significant among patients receiving SAE, crocin or placebo.

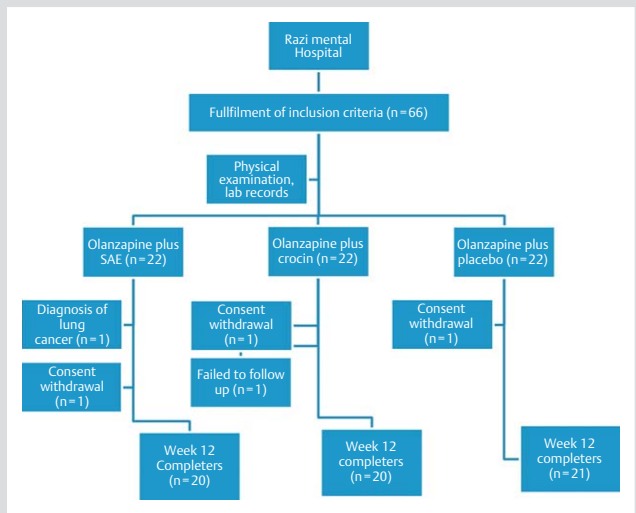

Fig. 1 Flow diagram of the enrolled patients with schizophrenia receiving olanzapine. (Color figure available online only).

Table 1 Characteristics of patients with schizophrenia assigned to each group.

	SAE Mean (SD)	Crocin Mean (SD)	Placebo Mean (SD)	p
age (year)	49.3 (7.1)	48.1 (7.7)	48.1 (6.1)	0.811
olanzapine (mg/day)	12.7 (4.3)	13.2 (3.9)	13.2 (3.3)	0.818
educational level (year)	7.7 (2.9)	8.4 (4.3)	8.1 (2.6)	0.904
duration of morbidity (years)	17.8 (5.6)	18.2 (6.2)	17.4 (5.3)	0.895

SAE = saffron aqueous extract, SD = standard deviation

Discussion

Patients with severe mental illness are more likely to have hyperglycemia/diabetes, hypertension and dyslipidemia [6, 29]. This risk has increased further by using atypical antipsychotics, especially olanzapine [1, 3]. Until now several studies have been designed to diminish the additional load, which is induced by antipsychotics treatment, but there still remains unresolved issues [30–32].

Saffron was used throughout the ancient world as a multipurpose medicine [22]. There has been a recent surge in interest and research about it and its components during the past 2 dec-

Table 2 Components of metabolic syndrome and lipid profile measured during 12 weeks in patients treated with olanzapine.

		Baseline Mean (SD)	Week 2 Mean (SD)	Week 6 Mean (SD)	Week 12 Mean (SD)	p	Effect size
FBS (mg/dL)	SAE	90.6 (7.3)	93.5 (6.7)	98.7 (9.5)	100.5 (7.1)	**0.004**	0.174
	crocin	93.8 (4.4)	93.6 (6.5)	98.2 (7.5)	97.9 (6.9)		
	placebo	90.1 (6.4)	99.6 (7.3)	105.3 (7.9)	108.6 (10.1)		
TG (mg/dL)	SAE	99.9 (27.9)	111.5 (41.8)	107.1 (35.0)	85.0 (27.5)	0.723	0.011
	crocin	101.0 (55.6)	103.7 (47.9)	125.1 (59.5)	102.8 (47.1)		
	placebo	109.7 (37.3)	112.6 (37.7)	112.1 (42.4)	107.3 (49.1)		
cholesterol (mg/dL)	SAE	168.6 (17.9)	171.7 (22.4)	182.6 (25.6)	168.5 (24.3)	0.598	0.018
	crocin	181.5 (34.2)	172.4 (32.0)	182.2 (32.5)	178.9 (34.4)		
	placebo	174.6 (26.4)	177.5 (35.4)	191.9 (34.2)	181.7 (34.4)		
LDL (mg/dL)	SAE	102.7 (17.6)	107.0 (20.2)	117.5 (20.7)	105.8 (23.3)	0.527	0.022
	crocin	111.2 (32.4)	106.7 (25.9)	114.8 (30.3)	112.5 (28.3)		
	placebo	108.0 (23.2)	115.3 (23.9)	125.9 (27.4)	115.8 (26.8)		
HDL (mg/dL)	SAE	46.0 (8.7)	45.3 (8.4)	47.6 (7.2)	52.8 (7.1)	0.109	0.073
	crocin	45.6 (9.7)	46.4 (11.9)	48.1 (11.9)	51.3 (9.7)		
	placebo	44.5 (13.6)	40.8 (11.2)	43.0 (13.5)	45.4 (11.5)		
waist circumference (cm)	SAE	91.2 (6.8)	–	91.6 (7.3)	92.1 (7.4)	0.186	0.056
	crocin	90.8 (7.7)	–	91.5 (8.6)	91.9 (8.6)		
	Placebo	92.6 (7.8)	–	95.3 (7.8)	98.4 (8.4)		
blood pressure	SAE	114.7 (5.9)	–	114.7 (5.9)	116.5 (5.6)	0.065	0.090
	crocin	111.2 (7.9)	–	111.5 (7.9)	109.7 (6.8)		
	placebo	111.5 (7.7)	–	114.5 (8.0)	115.9 (9.0)		

SAE: saffron aqueous extract, SD = standard deviation, FBS = fasting blood sugar, TG = triglycerides, LDL = low-density lipoprotein, HDL = high-density lipoprotein

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Table 3 Measured components of insulin resistance during 12 weeks in patients treated with olanzapine.

		Baseline Mean (SD)	Week 2	Week 6	Week 12 Mean (SD)	p	Effect size
fasting insulin level (mIU/mL)	SAE	4.0 (.9)	–	–	2.8 (1.3)	0.230	0.504
	crocin	4.6 (2.5)	–	–	4.0 (2.0)		
	placebo	4.4 (2.0)	–	–	3.5 (2.1)		
HbA1c (%)	SAE	5.0 (0.3)	–	–	4.8 (0.4)	0.419	0.030
	crocin	5.1 (0.3)	–	–	4.9 (0.5)		
	placebo	4.9 (0.3)	–	–	4.8 (0.4)		
HOMA-IR	SAE	917.0 (222.6)	–	–	740.9 (346.6)	0.199	0.054
	crocin	1114.1 (511.6)	–	–	963. 5 (484.4)		
	placebo	989.4 (475.2)	–	–	944.5 (592.5)		
HOMA β (%)	SAE	58.5 (23.6)	–	–	29.1 (12.8)	0.572	0.019
	crocin	56.1 (28.4)	–	–	43.1 (24.5)		
	placebo	60.9 (28.5)	–	–	29.0 (17.4)		

SAE = saffron aqueous extract, SD = standard deviation

ades because of the various biological and clinical properties, the subject that has been extensively reviewed by us [23]. One of the clinical properties of saffron is improving lipid profile; e.g., a recent report showed hypolipidemic, hypoglycemic, and improving insulin resistance effects of SAE and crocin in the streptozotocin-induced type 2 diabetic rats [17, 19]. But to the best of our knowledge, there has been no report of protective effects of SAE and its active constituent crocin on olanzapine-induced metabolic syndrome in patients with schizophrenia. Thus, in this triple blind placebo controlled clinical trial, we evaluated the effect of the mentioned compounds against olanzapine-induced hyperglycemia, dyslipidemia, and insulin resistance in patients with schizophrenia.

In the present study, both SAE and crocin were well tolerated by the patients and did not interfere with clinical improvement; besides, no side effects were observed. This was compatible with the previous reports on the safety of these herbal preparations in humans [27, 33].

As the results show, SAE could significantly mitigate olanzapine-induced metabolic syndrome compared to placebo. Although the number of patients who developed metabolic syndrome in the crocin-treated group was lower than that in the placebo group, this difference was not significant. ▶ **Table 2** also shows that SAE and crocin could also prevent an increase in FBS during the 12 weeks of treatment. A similar hypoglycemic effect of saffron has been reported previously in the rat model of diabetes [17, 19].

Although the level of triglycerides and LDL did not significantly change in either of these groups, the mean level of HDL increased above 50 mg/dL in patients receiving SAE and crocin as early as week 6, which indicates a reduced risk of metabolic syndrome. In the animal model of diabetes, both SAE and crocin could improve lipid profile, increase HDL and decrease triglycerides, LDL and total cholesterol, after 5 months of treatment [17, 19].

Metabolic effects of antipsychotics may be due to the increased appetite and subsequent weight gain, according to agonistic effects on histamine 1 and muscarinic acetylcholine receptors [8, 34]. The inhibitory effect of saffron extract on histamine (H1) receptors has been shown in guinea pigs [35]. Despite the fact that the increase in waist circumference was not statistically significant between the 3 groups, the number of patients who reached the threshold of having waist circumference as high as the criteria for metabolic syndrome was significantly lower in patients receiving SAE. Considering the trend of this increase, a longer duration of follow-up might reveal a significant difference in this regard as well.

It has been reported that SAE, crocin and crocetin increase insulin secretion in rat models of diabetes mellitus and insulin resistance [17, 19, 36]. However, in the current study, serum levels of fasting insulin, HOMA-IR and HOMA-β indices were not influenced by SAE or crocin. The level of HbA1c also showed no significant change.

Crocin is the active component of saffron and is a predominant carotenoid in the saffron stigma. But although we used similar amounts of SAE and crocin in the present study, the results did not indicate more effectiveness of crocin in preventing FBS increase. So we hypothesized that crocin was not absorbed as well as SAE from the intestine of the patients. Previous animal studies have also shown that crocin does not absorb well from GI when administered orally to mice [37] and rat [38]. Therefore, in an animal study, crocin was administered through intraperitoneal injection and its effectiveness was shown [19]. According to our searches, there are no pharmacokinetic studies of SAE and crocin in humans, except one that detected crocetin after 2 h of saffron tea administration in the serum of 4 normal human volunteers [39]. Clearly, our hypothesis needs further investigation.

Researchers have studied other agents to prevent or treat olanzapine-induced metabolic disturbances. Metformin is one of the most famous agents, but results are limited and not the same in all of the studies. The most consisting results are about its effect on weight gain [40]. There are reports indicating that metformin may safely assist olanzapine-treated patients in body weight, but cannot improve the lipid profile and the HbA1c levels [41]. On the other hand, some trials report that it could significantly reverse the increase in the body weight, fasting blood glucose, insulin secretion and insulin resistance, after 8-weeks of administration in half of the participants [32].

Another agent is topiramate, for which the most consisting results are about reducing weight gain in trials [42]. Furthermore, a decrease in glucose, cholesterol, triglyceride levels and systolic and diastolic blood pressure is also reported [43]. A recent review has gathered the limited data about these 2 and concluded that metformin results in greater weight loss than topiramate, while topiramate is associated with more risks [44]. This review recommended that lifestyle and dietary changes should be considered first. A treatment regimen was indeed chosen based on the benefits and assets for each patient. Considering the possible adverse effects of these 2 agents, especially topiratame [45], SAE might be a suitable substitute for patients experiencing side effects or willing to consume herbal medicines.

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In conclusion, this trial was the first report on the effectiveness of 30 mg daily SAE for preventing metabolic syndrome in patients with schizophrenia who were treated with olanzapine. This effect appeared as early as week 6. SAE and crocin were both well tolerated and could prevent an increase in FBS as well.

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Conflict of Interest

The authors have no conflict of interest to declare. This manuscript contains original unpublished work – not submitted for publication elsewhere.

References

1 *Hert M, Schreurs V, Vancampfort D et al.* Metabolic syndrome in people with schizophrenia: a review. World Psychiatry 2009; 8: 15–22
2 *Goff DC, Sullivan LM, McEvoy JP et al.* A comparison of ten-year cardiac risk estimates in schizophrenia patients from the CATIE study and matched controls. Schizoph Res 2005; 80: 45–53
3 *Meyer J, Koro CE, Italien GJ.* The metabolic syndrome and schizophrenia: a review. Int Rev Psychiatry 2005; 17: 173–180
4 *Hennekens CH, Hennekens AR, Hollar D et al.* Schizophrenia and increased risks of cardiovascular disease. Am Heart J 2005; 150: 1115–1121
5 *Baptista T, Kin NM, Beaulieu S et al.* obesity and related metabolic abnormalities during antipsychotic drug administration: mechanisms, management and research perspectives. pharmacopsychiatry 2002; 35: 205–219
6 *Newcomer JW.* Second-generation [atypical] antipsychotics and metabolic effects. CNS Drugs 2005; 19: 1–93
7 *Tschoner A, Engl J, Laimer M et al.* Metabolic side effects of antipsychotic medication. Int J Clin Pract 2007; 61: 1356–1370
8 *Nasrallah H.* Atypical antipsychotic-induced metabolic side effects: insights from receptor-binding profiles. Mol Psychiatry 2008; 13: 27–35
9 *Scheen A, De Hert M.* Drug-induced diabetes mellitus: the example of atypical antipsychotics. Rev médic Liège 2005; 60: 455
10 *Scheen A, De Hert M.* Abnormal glucose metabolism in patients treated with antipsychotics. Diab Metab 2007; 33: 169–175
11 *Van Winkel R, De Hert M, Wampers M et al.* Major changes in glucose metabolism, including new-onset diabetes, within 3 months after initiation of or switch to atypical antipsychotic medication in patients with schizophrenia and schizoaffective disorder. J Clin Psychiatry 2008; 69: 472
12 *Meyer JM, Pandina G, Bossie CA et al.* Effects of switching from olanzapine to risperidone on the prevalence of the metabolic syndrome in overweight or obese patients with schizophrenia or schizoaffective disorder: analysis of a multicenter, rater-blinded, open-label study. Clin Therapeut 2005; 27: 1930–1941
13 *Newcomer JW.* Antipsychotic medications: metabolic and cardiovascular risk. J Clin Psychiatry 2007; 68 (Suppl 4): 8–13
14 *Conn PJ, Lindsley CW, Jones CK.* Activation of metabotropic glutamate receptors as a novel approach for the treatment of schizophrenia. Trends Pharmacol Sci 2009; 30: 25–31
15 *Mauri M, Simoncini M, Castrogiovanni S et al.* A psychoeducational program for weight loss in patients who have experienced weight gain during antipsychotic treatment with olanzapine. Pharmacopsychiatry 2008; 41: 17–23
16 *Kristiansen LV, Huerta I, Beneyto M et al.* NMDA receptors and schizophrenia. Curr Opin Pharmacol 2007; 7: 48–55
17 *Shirali S, Bathaie SZ, Nakhjavani M et al.* Effects of saffron [Crocus sativus L.] aqueous extract on serum biochemical factors streptozotocin-induced diabetic rats. Iran J Med Arom Plants 2012; 28: 293–308
18 *Ghahghaei A, Bathaie SZ, Kheirkhah H et al.* The protective effect of crocin on the amyloid fibril formation of AB42 peptide in vitro. Cell Mol Biol Lett 2013; 1–12
19 *Shirali S, Zahra Bathaie S, Nakhjavani M.* Effect of crocin on the insulin resistance and lipid profile of streptozotocin-induced diabetic rats. Phytother Res 2013; 27: 1042–1047
20 *Akhondzadeh Basti A, Moshiri E, Noorbala AA et al.* Comparison of petal of Crocus sativus L. and fluoxetine in the treatment of depressed outpatients: a pilot double-blind randomized trial. Prog Neuropsychopharmacol Biol Psychiatry 2007; 31: 439–442
21 *Akhondzadeh S, Sabet MS, Harirchian MH et al.* Saffron in the treatment of patients with mild to moderate Alzheimer's disease: a 16-week, randomized and placebo-controlled trial. J Clin Pharm Ther 2010; 35: 581–588
22 *Mousavi SZ, Bathaie SZ.* Historical uses of saffron: Identifying potential new avenues for modern research. Avicenna J Phytomed 2011; 1: 57–66
23 *Bathaie SZ, Mousavi SZ.* New applications and mechanisms of action of saffron and its important ingredients. Crit Rev Food Sci Nutr 2010; 50: 761–786
24 *American Psychiatric Association.* Diagnostic and statistical manual of mental disorders: DSM-IV-TR®. American Psychiatric Pub 2000
25 *Turner R, Levy J, Rudenski A et al.* Measurement of insulin resistance and beta-cell function: the HOMA and CIGMA approach. Front Diab 1993; 12: 66–75
26 *National Institute of Health.* Third report of the national cholesterol education program expert panel on detection, evaluation, and treatment of high blood cholesterol in adults [Adult Treatment Panel III]. NIH Publ 2001; 1: 3670
27 *Modaghegh MH, Shahabian M, Esmaeili HA et al.* Safety evaluation of saffron [Crocus sativus] tablets in healthy volunteers. Phytomedicine 2008; 15: 1032–1037
28 *Cohen L, Manion L, Morrison K.* Research methods in education. 6th Edition. London, Routledge 2011
29 *Druss BG, Bradford WD, Rosenheck RA et al.* Quality of medical care and excess mortality in older patients with mental disorders. Arch Gen Psychiatry 2001; 58: 565
30 *Richards M, Chiba S, Ninomiya M et al.* Inhibition of olanzapine-induced weight gain by the retinoid analog AM-80. Pharmacopsychiatry 2013; 46: 267–273
31 *Anderson G, Maes M.* Melatonin: an overlooked factor in schizophrenia and in the inhibition of anti-psychotic side effects. Metab Brain Dis 2012; 27: 113–119
32 *Chen C-H, Chiu C-C, Huang M-C et al.* Metformin for metabolic dysregulation in schizophrenic patients treated with olanzapine. Progr Neuro-Psychopharmacol Biol Psychiatry 2008; 32: 925–931
33 *Mohamadpour AH, Ayati Z, Parizadeh MR et al.* Safety evaluation of crocin [a constituent of saffron] tablets in healthy volunteers. Iran J Basic Med Sci 2013; 16: 39
34 *Shahmansouri N, Farokhnia M, Abbasi S-H et al.* A randomized, double-blind, clinical trial comparing the efficacy and safety of *Crocus sativus* L. with fluoxetine for improving mild to moderate depression in post percutaneous coronary intervention patients. J Affect Disord 2014; 155: 216–222
35 *Boskabady MH, Ghasemzadeh Rahbardar M, Nemati H et al.* Inhibitory effect of Crocus sativus [saffron] on histamine [H1] receptors of guinea pig tracheal chains. Pharmazie 2010; 65: 300–305
36 *Xi L, Qian ZY, Shen XC et al.* Crocetin prevents dexamethasone-induced insulin resistance in rats. Planta Med 2005; 71: 917–922
37 *Asai A, Nakano T, Takahashi M et al.* Orally administered crocetin and crocins are absorbed into blood plasma as crocetin and its glucuronide conjugates in mice. J Agric Food Chem 2005; 53: 7302–7306
38 *Xi L, Qian Z, Du P et al.* Pharmacokinetic properties of crocin [crocetin digentiobiose ester] following oral administration in rats. Phytomedicine 2007; 14: 633–636
39 *Chryssanthi DG, Lamari FN, Georgakopoulos CD et al.* A new validated SPE-HPLC method for monitoring crocetin in human plasma – application after saffron tea consumption. J Pharm Biomed Anal 2011; 55: 563–588
40 *Praharaj SK, Jana AK, Goyal N et al.* Metformin for olanzapine-induced weight gain: a systematic review and meta-analysis. Br J Clin Pharmacol 2011; 71: 377–382

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41 *Baptista T*, *Rangel N*, *Fernández V et al.* Metformin as an adjunctive treatment to control body weight and metabolic dysfunction during olanzapine administration: A multicentric, double-blind, placebo-controlled trial. Schizoph Res 2007; 93: 99–108
42 *Wozniak J*, *Mick E*, *Waxmonsky J et al.* Comparison of open-label, 8-week trials of olanzapine monotherapy and topiramate augmentation of olanzapine for the treatment of pediatric bipolar disorder. J Child Adol Psychopharmacol 2009; 19: 539–545
43 *Narula PK*, *Rehan HS*, *Unni KES et al.* Topiramate for prevention of olanzapine associated weight gain and metabolic dysfunction in schizophrenia: A double-blind, placebo-controlled trial. Schizoph Res 2010; 118: 218–223
44 *Ellinger LK*, *Ipema HJ*, *Stachnik JM*. Efficacy of metformin and topiramate in prevention and treatment of second-generation antipsychotic-induced weight gain. Ann Pharmacother 2010; 44: 668–679
45 *Kanner AM*, *Wuu J*, *Faught E et al.* A past psychiatric history may be a risk factor for topiramate-related psychiatric and cognitive adverse events. Epilep Behav 2003; 4: 548–552

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